



AMERICAN INTERNATIONAL UNIVERSITY- BANGLADESH (AIUB)

Dept. of Computer Science

Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section: M

**Project Report on
Grocery Store Management System**

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Grocery Store Management System

Introduction

In modern times, online grocery shopping has become an essential part of daily life. Customers expect a fast and simple way to browse products, add items to a cart, and place orders from home. At the same time, store administrators need efficient tools to manage inventory, monitor sales, and handle user activities.

This project presents a **Grocery Store Management System** developed using C# Windows Forms. The system provides a complete platform where customers can shop online, admins can manage products and inventory, and super admins can control users and system operations. The goal is to create a simple, user-friendly, and efficient system that reflects a real-world grocery shopping environment.

Objectives

The main objectives of this system are:

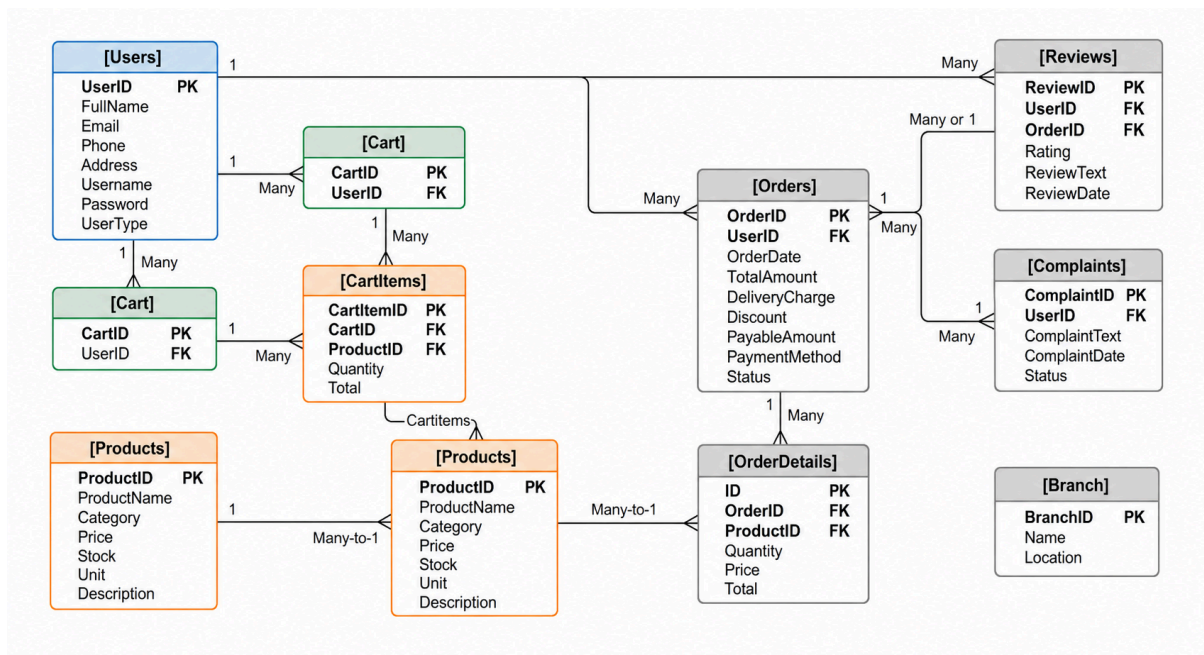
- To provide an easy-to-use platform for online grocery shopping
- To manage inventory efficiently
- To maintain customer records securely
- To automate order processing
- To reduce manual errors and improve accuracy

Case Study

This system models a real-world grocery shopping platform involving three types of users: Customer, Admin, and Super Admin. All users are stored in a single system, and their roles are determined automatically based on the stored user type. A customer named John begins by registering in the system, providing details such as full name, email, phone number, address, username, and password. After registration, he logs in using his credentials. The system identifies him as a customer based on his stored role. Once logged in, John accesses the Customer Dashboard. From there, he can navigate through sections such as Home, View Products, My Cart, My Orders, Profile, and Logout. In the product section, he browses grocery items like rice, sugar, cooking oil, and pulses. Each product

includes details such as product ID, category, price, available stock, unit, and description. John selects products and adds them to his cart. The system maintains his cart and allows multiple items to be added. In the Cart section, he can review selected items, update quantities, remove products, and view the total cost. When ready, John proceeds to checkout. He confirms his delivery address, and the system generates an order summary including total amount, delivery charge, discount, and final payable amount. He selects a payment method, such as Cash on Delivery, and places the order. The system then creates an order record along with detailed order items. Each order is associated with a specific branch responsible for fulfillment. John can track his order in the My Orders section, where he can view order ID, date, total amount, and current status such as Processing, Delivered, or Cancelled. After receiving the order, John can submit a review based on his experience. This feedback is stored and linked to his order, allowing administrators to evaluate service quality. Additionally, John can submit complaints if he faces any issues. These complaints are recorded and monitored by the system. The Admin manages daily operations through an administrative dashboard. They can add, update, or remove products, monitor stock levels, and ensure product availability. The dashboard also provides insights such as total orders, sales performance, and low-stock alerts, which are derived from system data. The Super Admin has full control over the system. They manage user accounts by blocking or unblocking users, oversee admin activities, add and manage branch information, and review customer feedback and complaints to ensure overall system quality. This system creates an efficient workflow where customers can shop conveniently, administrators can manage operations effectively, and continuous improvement is achieved through user feedback and system monitoring.

System Design (Database Schema)



Database Query

DATABASE

```
CREATE DATABASE GroceryStore;  
USE GroceryStore;
```

USERS

```
CREATE TABLE Users (  
    UserID INT PRIMARY KEY IDENTITY, FullName VARCHAR(100), Email VARCHAR(100),  
    Phone VARCHAR(20), Address VARCHAR(200), Username VARCHAR(50) UNIQUE,  
    Password VARCHAR(50), UserType VARCHAR(20)  
);
```

PRODUCTS

```
CREATE TABLE Products (  
    ProductID VARCHAR(10) PRIMARY KEY, ProductName VARCHAR(100),  
    Category VARCHAR(50), Price DECIMAL(10,2), Stock INT, Unit VARCHAR(20),  
    Description VARCHAR(255)  
);
```

BRANCH

```
CREATE TABLE Branch (  
    BranchID INT PRIMARY KEY IDENTITY, Name VARCHAR(100), Location VARCHAR(100)  
);
```

CART

```
CREATE TABLE Cart (  
    CartID INT PRIMARY KEY IDENTITY, UserID INT,  
    IsActive BIT DEFAULT 1, FOREIGN KEY (UserID) REFERENCES Users(UserID)  
);
```

CART ITEMS

```
CREATE TABLE CartItems (  
    CartItemID INT PRIMARY KEY IDENTITY, CartID INT, ProductID VARCHAR(10),  
    Quantity INT, Total DECIMAL(10,2), FOREIGN KEY (CartID) REFERENCES Cart(CartID),  
    FOREIGN KEY (ProductID) REFERENCES Products(ProductID)  
);
```

ORDERS

```
CREATE TABLE Orders (  
    OrderID VARCHAR(10) PRIMARY KEY, UserID INT, BranchID INT, OrderDate DATE,  
    TotalAmount DECIMAL(10,2), DeliveryCharge DECIMAL(10,2), Discount DECIMAL(10,2),  
    PayableAmount DECIMAL(10,2), PaymentMethod VARCHAR(50), Status VARCHAR(50),  
    FOREIGN KEY (UserID) REFERENCES Users(UserID), FOREIGN KEY (BranchID)  
    REFERENCES Branch(BranchID)  
);
```

ORDER DETAILS

```
CREATE TABLE OrderDetails (  
    ID INT PRIMARY KEY IDENTITY, OrderID VARCHAR(10), ProductID VARCHAR(10),  
    Quantity INT, Price DECIMAL(10,2), Total DECIMAL(10,2),  
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),  
    FOREIGN KEY (ProductID) REFERENCES Products(ProductID)  
);
```

REVIEWS

```
CREATE TABLE Reviews (  
    ReviewID INT PRIMARY KEY IDENTITY, UserID INT, OrderID VARCHAR(10),  
    Rating DECIMAL(3,1), ReviewText VARCHAR(500), ReviewDate DATE,  
    FOREIGN KEY (UserID) REFERENCES Users(UserID),  
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)  
);
```

COMPLAINTS

```
CREATE TABLE Complaints (  
    ComplaintID INT PRIMARY KEY IDENTITY, UserID INT, ComplaintText VARCHAR(500),  
    ComplaintDate DATE, Status VARCHAR(50),  
    FOREIGN KEY (UserID) REFERENCES Users(UserID)  
);
```

Features and Facilities

1. User registration and secure login
2. Role-based access (Customer, Admin, Super Admin)
3. Product browsing and search
4. Detailed product information
5. Add to cart and manage cart items
6. Order placement with checkout system
7. Order tracking and history
8. Customer review and feedback system
9. Inventory management with stock alerts
10. Admin dashboard with analytics
11. User account control by super admin
12. Complaint management system

Conclusion

The Grocery Store Management System successfully provides a complete platform for online grocery shopping and management. It integrates customer interaction, order processing, and administrative control into a single system.

Customers can easily browse products, manage their cart, place orders, and track their purchases. The review system allows users to share feedback, helping improve service quality. Admins can efficiently

manage inventory and monitor system performance, while the super admin ensures proper control over users and system operations. Overall, the system demonstrates a practical and efficient implementation of a real-world grocery store application using C#, making the process organized, user-friendly, and effective for all users.

1. Login Form

2. Customer Registration Form

3. Customer Dashboard (Home Form)

4. Product List Form

Product ID	Product Name	Category	Price	Stock	Unit
P001	Rice (5kg)	Grains	450.00	50	Pack
P002	Wheat Flour (5kg)	Flour	300.00	40	Pack
P003	Sugar (1kg)	Grocery	60.00	100	Pack
P004	Cooking Oil (1L)	Oil	180.00	35	Bottle
P005	Toor Dal (1kg)	Pulses	120.00	60	Pack

5. Product Details Form



Product ID

P001

Product Name

Rice (5kg)

Category

Grains

Price

450.00

Stock

50

Unit

Pack

Description

Premium quality rice, best for daily use.

Add to Cart

Close

6. Cart Form

My Cart

My Cart

Product ID	Product Name	Price	Quantity	Total	
P001	Rice (5kg)	450.00	1	450.00	✗
P003	Sugar (1kg)	60.00	2	120.00	✗
P004	Cooking Oil (1L)	180.00	1	180.00	✗

Grand Total:

750.00

Clear Cart

Place Order

Close

7. Checkout / Place Order Form

Checkout

Delivery Address

☒ Use Default Address

☐ Use New Address

John Doe

123, Green Street,

New City, NY - 10001

9876543210

Change Address

Order Summary

Total Items	3
Total Amount	750.00
Delivery Charges	50.00
Discount	0.00
Payable Amount	800.00

Payment Method

Cash on Delivery

Place Order

Cancel

8. My Orders Form

My Orders

Order ID	Order Date	Total Amount	Status	Details
ORD001	01-05-2024	750.00	Delivered	<button>View</button>
ORD002	05-05-2024	620.00	Shipped	<button>View</button>
ORD003	10-05-2024	330.00	Processing	<button>View</button>
ORD004	15-05-2024	910.00	Cancelled	<button>View</button>
ORD005	20-05-2024	450.00	Delivered	<button>View</button>

Close

9. Profile Form

My Profile



My Profile

Full Name

John Doe

Email

john.doe@email.com

Phone Number

9876543210

Address

123, Green Street,
New City, NY - 10001

Update

Cancel

ADMIN: Inventory Management

ADMIN: Inventory Management

Inventory

DataGridView

ID	Name	Category	Price	Stock
307	Shacall Name	Category	\$20.00	\$0
302	Ramakhichnent	Category	\$20.00	30
303	Eseonshont Name	Category	\$50.00	30
208	Low Stock	Low Stock	\$18.00	50
205	Product Riihan	Category	\$10.00	50
205	Low Stock	Low Stock	\$18.00	50
203	Rechine Name	Category	\$20.00	40
207	Beoli Name	Category	\$10.00	30
208	Kety Rich	Category	\$20.00	40
209	Retimir Name	Category	\$25.00	30

Add New Product

Update

Delete

Stock Alert

ADMIN: Dashboard

FormAdminDash

Dashboard

Professional Overview

Total Sales

\$182.3K

Total Orders

1,995

Low Stock Items

12

Total Customers

103

Monthly Sales Trend



Recent Orders

Recent Orders (2,000)

Jansruary 14, 2021

\$1,000

Recent Orders (2,000)

February 14, 2021

\$1,000

Recent Branch/ta...

Jansruary 14, 2021

\$1,000

Recent Customers

February 14, 2021

\$1,000

SUPER ADMIN: User & Branch Control

SUPER ADMIN: User & Branch Control

User & Accounts

All Registered Admin

Admin	Branch
Admindamis.com	Admin
CholinaEvracbaidd.com	Admin
Nasshpodaziat.com	Customer
Raranhwn.com	Admin
Nagel Admin	Customer

Block/Unblock

Block/Unblock

Block/Unblock

Add New Branch

System Reviews & Complaints

Date	Reviews	Complaint
2/19/2023 9:13 PM	Templation aswents	Grocery Express
2/18/2023 9:37 PM	Renesttows & Complaints	Grocery Express
2/10/2023 5:17 PM	System Reviews & Complaints	Grocery Express
2/10/2023 9:19 PM	Templation aswents	Grocery Express